MIL-P-21386E(EC)
7 June 1985
SUPERSEDING
MIL-P-21386D(SHIPS)
8 May 1967

MILITARY SPECIFICATION

PAPER, FILTER MEDIA, ROLL-TYPE

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers filter paper for use in air particle radiation monitors, hereinafter referred to as the filter paper.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-P-31	Paper, General Specifications And Methods Of Testing
CCC-T-191	Textile Test Methods
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-676	Boxes, Setup
PPP-T-60	Tape, Pressure Sensitive, Adhesive, Waterproof - For Packaging And Sealing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (For Carton Sealing)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-129	Marking For Shipment And Storage

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

TAPPI 411	Thickness (Caliper) Of Paper And Paper Board

(Application for copies should be addressed to Technical Association of the Pulp and Paper Industry (TAPPI) 15, Technology Park S/Box 105113, Atlanta, GA 105113)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR 8111), Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6640

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI PH1.33.80 Spool, 16 mm, 100 Foot, 16 mm, 200 Foot, 35 mm, 100 Foot And 70 mm, 100 Foot, For Recording Instruments And For Microfilm And Still Cameras, Dimensions For

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The filter paper shall be designed such that with an air flow of 2 cubic feet per minute (cfm) to 4 cfm per square inch (in.) of the filter paper, the filter paper shall collect particles of 0.3 micrometer (µm), or larger, in diameter with at least 98 percent efficiency.

3.2 Material. The material used in the manufacture of the filter paper shall be subject to procuring activity approval.

3.3 Physical characteristics. Physical characteristics of the filter paper shall be as specified in TABLE I.

TABLE I. Filter paper physical characteristics.

Characteristics	Requirements
Thickness	3.6 mil ±0.5 mil
Width	1.25 in. ±0.03 in.
Length	100 ft 1/ ±5 ft; no splices permitted
Resistance to passage of air at a low rate of volume:	
For 0.5 in. of water	Average shall fall within range of 4.5 cfm to 9 cfm per square ft of paper, with a maximum variation from average of ±0.5 cfm
For 1 in. of water	Average shall fall within range of 9.0 cfm to 17 cfm, with a maximum variation of ±0.8 cfm
Resistance to passage of air at a high rate of volume	Average shall fall within a range of 400 mm 2/ to 800 mm of water with a maximum variation of ±25 mm of water from the average
Percent efficiency	Not less than 98 percent in any sample sheet for 0.3 µm, or larger particles
Tensile strength	Minimum of 8.5 pounds per linear in.

1/ Feet
2/ Millimeter

3.4 Appearance. The filter paper shall be clean, free of holes and dirt spots, and comparatively free of bulked tufts of loose fiber on the surface. The filter paper shall be free of fold-over, creases, damaged edges, and other functional damages.

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3.5 Spools. The spools on which the filter paper is wound shall be plastic or metal type with radial slot, and shall conform to ANSI PH. 1.33-1980.

3.5.1 Winding. Filter paper on rolls shall be anchored by use of the reel slot; no tape will be permitted. The filter paper shall be wound with the reinforcement on the outside.

3.6 Visual. The filter paper shall be wound in a clockwise direction when facing the keyway side of the reel. Even winding shall be maintained throughout the length of the roll without fold-over, (ridges are not defects) damaged edges, or other damages. The outside end of the roll of filter paper shall be anchored to the roll with pressure-sensitive tape. Care shall be taken to ensure that the filter paper is not wound so tight as to be damaged in any way.

3.7 Markings. A printed label bearing the following inscription in block type shall be securely affixed to the outside of the container:

6640-01-188-5418
PAPER, FILTER
ROLL-TYPE, 100 ft
1 Roll
(date packaged)
Lot No.

FOR AIR PARTICLE
RADIATION MONITORS

3.7.1 Type size. The first three lines specified in 3.7 shall be considerably larger and bolder than the remainder of the inscription on the label. Special care shall be taken to use an adhesive that will ensure permanent attachment of the label to the container. Stock number, manufacturer's name, batch number or paper base, roll number, and date packaged shall be inserted in the spaces indicated. The letter size used for inserting manufacturer's name shall not exceed the smallest letter size used for the remainder of the labels.

3.8 Workmanship. The quality of workmanship shall promote paper filter conformance to the requirements specified in 3.1 and 3.7.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Sampling. Sampling shall be as specified in 4.2.1 through 4.2.3.2.

4.2.1 Lot. For purposes of sampling, a lot shall consist of all the rolls of filter paper which are made from the same batch of stock.

4.2.1.1 Shipment lot. Shipment lot shall consist of the quantity in one shipment to a depot.

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4.2.2 Sampling for examination. From each shipment lot, sample rolls shall be randomly selected in accordance with MIL-STD-105 at inspection level II and acceptable quality level 2.5 percent defective, to determine compliance with 4.3.

4.2.3 Sampling for tests. The filter paper stock shall be randomly sampled on a footage basis, or by other means approved by the procuring activity. For footage-basis sampling, 2 ft shall be removed from each randomly selected roll.

4.2.3.1 Resistance to passage of air at a low rate of volume. Sampling shall be in accordance with level S-4 of MIL-STD-105, except that the number of specimens shall be not less than six and need not exceed 15. The acceptance level for individual values failing to conform to the airflow tolerances specified in 3.3 shall be based on 2.5 percent defective. The tests shall be in accordance with 4.4.2.

4.2.3.2 Resistance to passage of air at a high rate of volume and filtering efficiency. When tests for resistance to passage of air at a high rate of volume and filtering efficiency are specified (see 6.2), sampling shall be in accordance with level S-4 of MIL-STD-105. The number of samples for each test need not exceed six. The acceptance level for individual values failing to conform to the resistance and percent efficiency tolerances specified for the applicable requirement shall be based on 2.5 percent defective. When tests in accordance with 4.4.3 and 4.4.4 are specified, the tests shall be conducted at a laboratory satisfactory to the procuring activity.

4.3 Visual and dimensional examination. The filter paper, sampled in accordance with 4.2.2, shall be examined to determine compliance with the visual and dimensional requirements of this specification. At least 3 ft of material shall be unrolled from each sample roll to provide an adequate sample (to ease rerolling on spool).

4.4 Tests for physical characteristics. Tests for physical characteristics shall be as specified in 4.4.1 through 4.4.5.

4.4.1 Thickness. Thickness of filter paper shall be determined in accordance with Method 173 of UU-P-31 (3.6 mil thickness) or TAPPI 411, except that the presser foot shall be circular with a diameter of 0.625 in. and with the moving parts connected therewith, dead weighted so as to apply a total load of 7 pounds per square inch, at a controlled rate of applied pressure.

4.4.2 Resistance to passage of air at a low rate of volume. The resistance of the filter paper to airflow at a low rate of volume shall be conducted in accordance with Method 5450 of CCC-T-191, except that airflow shall be measured at resistance values of 0.5 in. and 1.0 in. of water, respectively.

4.4.3 Resistance to passage of air at a high rate of volume. The apparatus for this test shall be such that the paper can be suitably clamped over an orifice exposing 1.86 square centimeters (cm^2) of filter paper. Air shall be accurately regulated to flow through this area of the filter paper at 16 liters per minute. The apparatus shall be equipped to measure accurately the resistance of the filter paper to the airflow. The resistance of the filter paper to airflow shall be reported in mm of water.

4.4.4 Filtering efficiency. The apparatus for this test shall be such that the filter paper can be suitably clamped over an orifice exposing 1.86 cm^2 of filter paper. The apparatus shall be such that it can generate particles 0.3 µm in diameter which are carried in a stream of air flowing at a rate of 16 liters per minute through the exposed 1.86 cm^2 area of the filter paper. The apparatus shall have suitable means of accurately measuring the penetration of the particles into the filter paper. The results of the test shall be reported in percent efficiency or percent penetration.

4.4.5 Tensile strength. The filter paper shall be tested to determine compliance with the requirements of TABLE I.

4.5 Inspection of preparation for delivery. Inspection shall be performed to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

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5.1 Preservation and packaging. Preservation and packaging shall be in accordance with Level A or Level C, as specified (see 6.2).

5.1.1 Level A. Level A requirements shall be as specified in 5.1.1.1 and 5.1.1.1.1.

5.1.1.1 Unit packaging. Each roll shall be packaged in a round, snug-fitting metal container with a slipcover closure, approximately 4 in. diameter. Containers made of ferrous material shall be coated on the exterior with a rust-resisting coating. The container shall be sealed with waterproof pressure sensitive tape conforming to PPP-T-60, Class 1 or PPP-T-76.

5.1.1.1.1 Intermediate containers. Rolls, packaged as specified in 5.1.1.1, shall be placed in intermediate containers conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size and shape, shall be of minimum cube and tare, and shall contain multiples of five unit packages, not to exceed 100 unit packages or 10 pounds. No intermediate packaging is required when the total quantity shipped to a single destination is less than 100 units.

5.1.2 Level C. Each roll shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity. This package may conform to contractor's commercial practice for retail distribution when such conforms to the requirements of this level.

5.2 Packing. Packing shall be in accordance with Level A, Level B, or Level C as specified (see 6.2).

5.2.1 Level A. The packaged roll(s) shall be packed in fiberboard boxes conforming to PPP-B-636, class weather-resistant, style optional, special requirements use. In lieu of the closure and waterproofing requirements in the Appendix of PPP-B-636, closures and waterproofing shall be accomplished by sealing all seams, corners, and manufacturer's joints with waterproof tape, 2-in. minimum width, conforming to PPP-T-60, Class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the Appendix of PPP-B-636, using tape or non-metallic only.

5.2.2 Level B. The packaged roll(s) shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style and use requirements (see 6.2). Closures shall be in accordance with the appendix thereto.

5.2.3 Level C. The packaged roll(s) shall be packed in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. This package shall conform to the applicable carrier rules and regulations and may be contractor's commercial practice when such conforms to the requirements of this level.

5.3 Marking. In addition to the marking required by 3.7 and any special marking by the contract or purchase order, unit packages, intermediate, and exterior containers shall be marked in accordance with MIL-STD-129.

5.4 General. Exterior containers shall be of a minimum cube and tare consistent with the protection required and shall contain equal quantities of identical items to the greatest extent possible.

6. NOTES

6.1 Intended use. The filter paper covered by this specification is intended for use in air particle radiation monitors.

6.2 Ordering data. Procurement documents should specify:

a. Title, number, and date of this specification
b. Whether or not tests for resistance to passage of air at a high rate of volume and filtering efficiency will be required for lot acceptance (see 4.2.3 2)
c. Levels of packaging and packing required (see 5.1 and 5.2)
d. Class of box required (see 5.2.2)

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
NAVY-EC

(Project No. 6640-N085)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER: MIL-P-21386 E

2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional

b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional

c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional

8. DATE OF SUBMISSION *(YYMMDD)*

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